AutoGen

AutoGen is **an open-source framework developed by Microsoft Research for building applications that use multiple AI agents to collaborate and solve complex tasks**. It simplifies the process of creating these "agentic AI" systems by handling the coordination, communication, and memory between agents, allowing developers to focus on building and customizing them with less code. The framework uses an event-driven, asynchronous architecture and is particularly useful for tasks like code generation, data analysis, and automation.

Key features and functions

- **Multi-agent collaboration**: AutoGen enables multiple AI agents, powered by large language models (LLMs), to have structured conversations to work together on a problem.
- **Simplified development**: It abstracts away the complexity of managing communication, state, and memory between agents, which accelerates the development of complex applications.
- **Flexibility**: The framework offers both pre-built templates and the flexibility for advanced customization.
- **Extensibility**: It has a Python SDK for easy installation and integration, and allows for workflows to be exported as APIs for deployment.
- **Task-oriented**: It is designed for AI projects that involve tool usage, code generation, and automation.
- **Core components**: The architecture consists of a core layer for message passing and runtime, and an AgentChat layer for conversation orchestration.

Use cases

- Solving real-world problems like supply-chain optimization.
- Performing complex data analysis and content generation.
- Automating workflows that require multiple steps and decision-making.